C-HI Early Snapshot Series

Date: 2026-05-10

Summary

This report extends the aligned C-HI, 0 deg smoke lane from one snapshot to the five regular early science outputs already available from the particles-enabled kunanyi run:

Grid output	Time (Myr)	Particle output
2	0.9990	101
4	1.9996	202
6	2.9999	303
7	3.9991	403
8	4.9994	503

For each snapshot, the same compact lane was run:

- Level 1 injected-energy truth
- Level 2 tracer-defined lobe truth at tracer > 5e-3
- first-pass ellipsoidal cavity protocol
- XRAiSE-corrected 1 keV grid-thermal X-ray scalar smoke
- particle-loss-aware 150 MHz radio/lobe scalar smoke

The radio extraction used the per-snapshot Level 2 side volumes, not the previous fixed 5 Myr lobe volumes.

Kunanyi Job

- job id: 690646.kunanyi-ohpc
- job name: c_hi_eseries
- node: n001
- resources: select=1:ncpus=4:mpiprocs=1
- compact archive: /u/bwbrook/codex_kunanyi/c_hi_early_snapshot_series_2026-05-10.tar.gz
- local archive: build/science/kunanyi_stage_2026-05-08/c_hi_early_snapshot_series_690646_2026-05-10.tar.gz
- archive SHA256: 503da7d4d2e80f1a2a76e5946aee59724ffe5055a71893119ef6786a06d8b68e

The only stderr warning was the known `NumbaParallelSafetyWarning` from the supplied particle-loss path. The 5 Myr target for this same lane has already passed a `NUMBA_NUM_THREADS=1` versus `4` deterministic check.

Main Trends

Metric	1 Myr	5 Myr
bipolar injected energy	`6.3052e58 erg`	`3.1554e59 erg`
Level 2 tracer-lobe energy	`4.2795e58 erg`	`1.8459e59 erg`
bipolar `4pV` cavity proxy	`3.5998e58 erg`	`2.5367e59 erg`
Level 2 tracer-lobe volume	`539.62 kpc^3`	`10791.77 kpc^3`
tracer z extent	`10.08 kpc`	`33.04 kpc`
XRAiSE-corrected X-ray peak	`2.8772e6 mJy sr^-1`	`2.8783e6 mJy sr^-1`
lossy 150 MHz luminosity proxy	`1.2695e28 W Hz^-1`	`6.3450e27 W Hz^-1`
radio lobe balance, `+z / -z`	`1.0579`	`0.9851`

The early series is coherent:

- injected energy grows linearly, as expected for the still-active jet
- tracer volume grows by about `20x` from 1 to 5 Myr
- by 5 Myr, Level 2 tracer-lobe energy is about `0.585` of elapsed bipolar injection
- by 5 Myr, the bipolar `4pV` proxy is about `0.804` of elapsed bipolar injection
- radio lobe balance stays close to unity across the five snapshots, ranging from `0.9720` to `1.0579`
- the central X-ray peak is essentially flat across this early interval, while the particle-radio peak and luminosity evolve more strongly

Figure

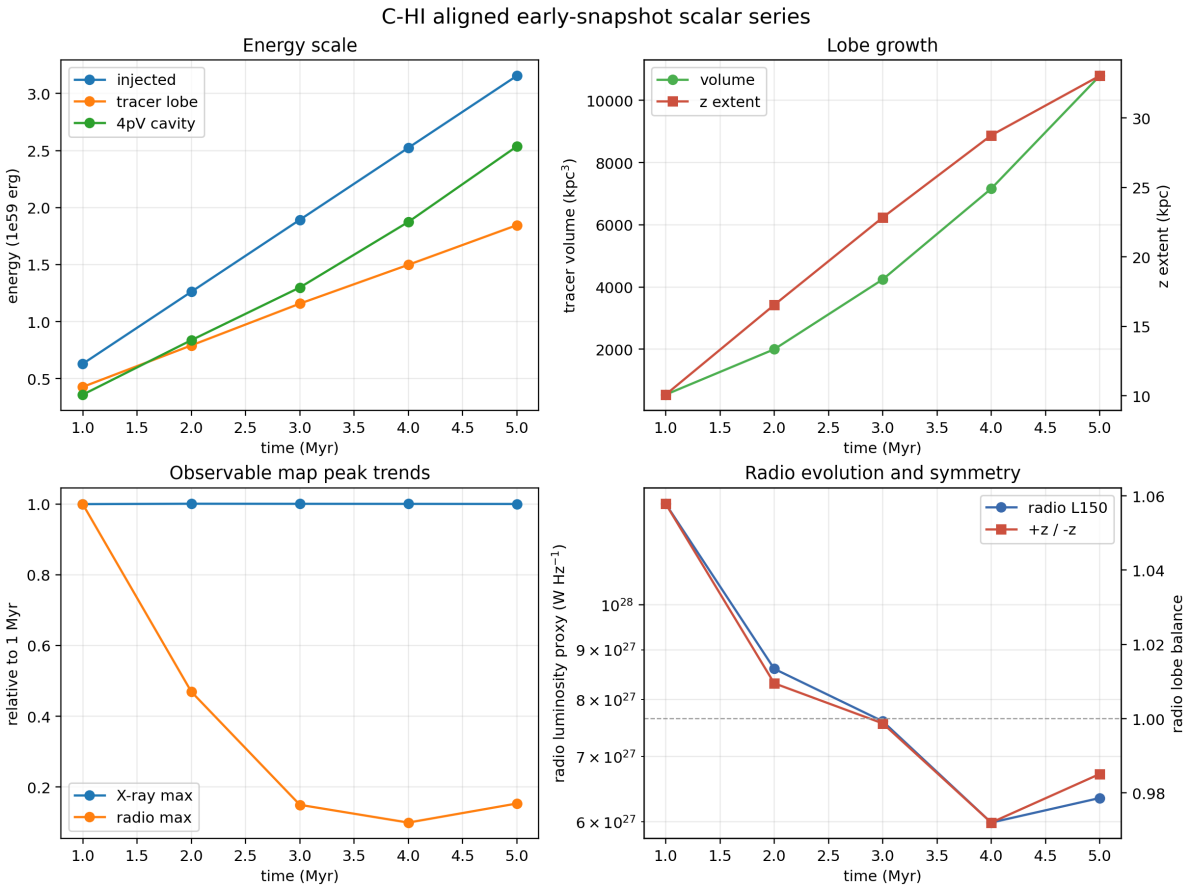


Figure 1: C-HI early snapshot scalar series

Products

Compact products are stored under:

```
docs/science/assets/c_hi_early_snapshot_series_2026-05-10/
```

Key files:

- `early_snapshot_series_metrics.csv`
- `c_hi_early_snapshot_series.png`
- `compact/snap_0002/`
- `compact/snap_0004/`

- compact/snap_0006/
- compact/snap_0007/
- compact/snap_0008/

Reusable scripts:

- scripts/kunanyi/c_hi_early_snapshot_series.pbs
- scripts/science/build_early_snapshot_series.py

Interpretation

This closes the early-snapshot scalar-series gate. The first five regular outputs show smooth lobe growth, stable symmetry, sensible energy ordering, and no immediate evidence from the compact diagnostics that the C-HI setup should be stopped before a longer continuation.

The next compute-facing move can therefore be a guarded continuation of C-HI toward the first 50 Myr milestone, using the already validated restart, -maxtime, output-catalogue, and email-notification pattern. The practical near-term target should be the next block of regular snapshots rather than a blind multi-day push all the way to completion.